

Palestine SME Social Media Intelligence Platform

Data Mining Final Project



Group Number:

1

Group Members:

1. Hana Izzdeen
2. Dareen Abu Aisheh
3. Mohammed Nabulsi
4. Hani Basha

Course:

Data Mining

Instructor:

Dr. Ahmad Alia

Date:

May 2026

Data Mining Final Project Submission Form

1. Group Information

Group Number:

1

Group Leader:

Mohammed Nabulsi

Project Title:

Palestine SME Social Media Intelligence Platform

No.	Group Member Full Name
1	Hana Izzdeen
2	Dareen Abu Aisheh
3	Mohammed Nabulsi
4	Hani Basha

2. Contribution and Main Tasks

Member Name	Contribution Percentage	Main Tasks (top 5)
Hana Izzdeen	25%	Time Series Analysis, Weekly Trend Mining, and Business Momentum Evaluation
		Explainable Anomaly Detection for Social Media Engagement
		Engagement Forecasting using Prophet Models
		Network Relationship and Engagement Pattern Analysis
Mohammed Nabulsi	25%	Dimensionality Reduction using t-SNE for clustering visualization
		BERTopic Clustering using BERT, UMAP, HDBSCAN, and c-TF-IDF
Dareen Abu Aisheh	25%	Similar Business Recommendation & Benchmarking System

		Created a Benchmark Analytics Dashboard
		Next Post Recommendation System
		Association Rule Mining
Hani Basha	25%	Used K-Means and DBSCAN for clustering
		Applied t-SNE for dimensionality reduction.
		Identified content-style patterns and similar post groups
All team members collaborated in collecting the dataset, contributing more than 700 rows in total, with each member collecting approximately 200 rows. Each member worked on their own project module, including both the backend and frontend interface implementation.		

3. Repository, Demo, and Presentation Links

GitHub Repository URL:

<https://github.com/Mohammad-Nabulsi/marketing.git>

Demo Video URL:

<https://github.com/Mohammad-Nabulsi/marketing/tree/0ac5f3398c6c45f403cc10f50af5053bf507932d/video>

Presentation File URL:

<https://github.com/Mohammad-Nabulsi/marketing/blob/2be18d58600dea1dc4d3cff11c5a816754fe8267/Palestine-SME-Social-Media-Intelligence-Platform.pdf>

4. Simple Abstract

Your abstract should simply answer these questions:

- What problem does your project solve?
- Why is this problem important?
- What data mining techniques did you use?
- What does your system/application do for the user?

- What are the main results or useful outputs?

Abstract:

This project aims to help Palestinian small and medium businesses understand and improve their social media performance using data mining techniques. Many businesses publish content online without clear insights about what improves engagement, what causes weak performance, or how their performance compares to competitors in the same sector. This problem is important because social media has become a major marketing channel for businesses. Our system analyzes social media datasets and provides useful insights through clustering, anomaly detection, forecasting, association rules, time series analysis, and network analysis. The platform helps users discover engagement trends, identify high-performing and weak posts, compare business performance with sector averages, forecast future engagement, and explore relationships between sectors, content types, and engagement levels. The project generates visual charts, recommendation outputs, anomaly reports, weekly trend analysis, and forecasting results that support better marketing decisions and content strategies for businesses.

5. Data Mining Techniques Used

Technique	Used? Yes/No	Algorithm / Method	Purpose in the Project
Clustering	Yes	K-Means , DBSCAN	Group posts and businesses based on engagement behavior and content patterns
Dimensionality Reduction	Yes	t-SNE	Reduce feature dimensions and visualize business/post patterns
Recommender Systems	Yes	KNN-based Similar Business Recommendation + Association-Rule-Based Recommendations	Generate content and engagement recommendations for businesses
Time Series Analysis	Yes	Weekly Trend Analysis, Business Momentum Analysis, Prophet Forecasting	Analyze engagement trends over time and predict future engagement performance
Network Analysis	Yes	Relationship Graph Analysis	Discover relationships between sectors, post types, languages, and engagement levels
Anomaly Detection	Yes	Isolation Forest, Local Outlier Factor (LOF), Z-Score	Detect sector-based positive and negative engagement anomalies and explain unusual business behavior
Association Rule Mining	Yes	Apriori Algorithm	Discover hidden relationships between engagement behaviors and content patterns

Other	Yes	Business Momentum Analysis	Compare business performance against sector averages and identify momentum trends.
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6. AI Generation Usage Declaration

AI Tool / Model	Purpose	Approx. Usage Ratio (%)
ChatGPT	Improve English, report writing, explanations, debugging assistance, and project documentation	10%
GitHub Copilot	Code suggestions, auto-completion, and implementation support	15%
Google Colab AI / VS Code AI Tools	Code testing and debugging assistance	5%

Total estimated AI-generated or AI-assisted contribution ratio: *Example: 15% of the whole project.*

7. Limitations

The project faced limitations related to limited historical data for some businesses, which affected forecasting and anomaly detection accuracy. Data collection was also challenging because some social media posts had missing or incomplete columns, which required manual completion and preprocessing. In addition, the analysis depends mainly on social media engagement metrics and does not include external business factors such as paid advertising or market conditions. Some forecasting and anomaly detection results may become less reliable for businesses or sectors with limited historical posting data.

8. Future Work

Future work includes integrating live social media APIs for automatic real-time data collection, improving forecasting accuracy with deeper historical data, and adding more explainable AI features for recommendations and anomaly analysis. The platform can also be expanded with interactive dashboards, business comparison tools, and personalized strategy suggestions for different sectors.

9. Special Contribution

The strongest feature of our project is the integration of multiple data mining techniques into one intelligent social media analysis platform for Palestinian SMEs. The system does not only visualize engagement data, but also provides practical business value through anomaly detection, forecasting, business-vs-sector comparison, recommendation generation, and network relationship analysis. In addition, The platform considers Arabic social media content and language-related features when available, which makes the analysis more relevant and practical for Palestinian businesses. The project combines clear visual insights with explainable outputs that help businesses understand their social media performance and improve their content strategies